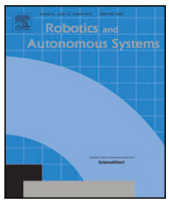
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A new SMART gripper with soft fingers and integrated force sensors for adaptive robotic tasks

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ABSTRACT

In the last decades, robotic manipulation has been treated as a cornerstone research topic to improve, accelerate but also safeguard human work in industrial, medical and everyday activities. For this reason, robotic hands are increasingly devoted to mimicking or augmenting human abilities. In particular, the two main operations performed by a human hand are precision and power grasping with the possibility of dosing the right grasping force to avoid breaks, deformations, or drops of the grasped objects. This paper presents the design, characterization and testing of a novel underactuated tendon-driven soft robotic gripper with sensing fingertips to regulate the applied contact force mimicking the ability of the human hand. The evolution of the design phases is presented and discussed together with the characterization procedures. Results of the grasping tests show the successful application of the new gripper in different tasks.

1. Introduction

In the last decades, important advancements have been focused on the development of industrial robots and their end-effectors, to help human operators in their tasks, especially in dangerous and repetitive operations. One of the most interesting human capabilities to be reproduced or augmented to help and protect human workers is the grasping ability and dexterity [1]. For this reason, many research activities are increasingly focused on the introduction and implementation of new efficient robotic hands to be used as end-effectors, or simply handled and managed by human operators [2–4]. Robotic grippers have the aim of handling, grasping and manipulating different kinds of objects through gripping organs that come into contact with the target, avoiding damage, breakage, or deformation, and performing positioning operations. The gripping organs can have different shapes and structures; they are often in the form of fingers, which allow grasping or supporting the object by opening or closing operations [5]. Grasping operations can be classified into two main categories, power and precision grasping, depending on the specific task to be accomplished. In particular, power grasping allows to wrap and sustain objects with higher contact forces and surfaces [6], while precision grasping is used to handle small and delicate objects that must be accurately manipulated and placed using fingertips [7]. This leads to different characteristics and structure of robotic grippers and fingers. Depending on the type of application, both the fingers and the gripper structure will have different features and can be classified according to different criteria such as the number and flexibility of the fingers, actuation, application, size, structure, and design [8]. Traditionally, robotic fingers consist of a series of rigid links and rigid or flexible joints [9,10], but recently, new advances have been focused on soft robotic solutions, such as anthropomorphic grippers, which are more compliant and mechanically simpler, using flexible fingers or gripping tools to better adapt to the object being grasped [11,12]. In numerous case studies, soft gripper prototypes have been developed by exploiting the elastic properties of different materials and different actuation techniques [13].

Soft robotic fingers can be made from various or combined materials such as Smart Memory Alloys and polymer materials [14], or silicone and nitinol [15]. Soft fingers can be structured as assemblies of different modules or can be conceived as monolithic structures [16,17]. Adhesive properties are also exploited and studied to create fingertips that mimic natural abilities [18].

Regarding actuation, soft robotic grippers can exploit pneumatic, hydraulic, or electric actuation by tendon-driven mechanisms [7,19,20]. Recently, several underactuated tendon-driven devices have been increasingly studied and developed [21,22]. In fact, the softness of the fingers and the underactuation of the mechanism can increase the compliance of grasping and the shape adaptability of the fingers, ensuring an effective grasping by tuning the finger stiffness and allowing an elastic return to the starting position [23–25].

In underactuated tendon-driven robotic grippers, a differential system is needed to allow effective and complete grasping operations

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even if one or more fingers get in touch with the object before the others [26]. One of the most used types of differential mechanism for tendon-driven robotic grippers is the pulley type [27,28]. In other cases, a set of gears is used to act as a differential mechanism [29]. Seesaw systems can also be used for the transmission of forces in underactuated tendon-driven grippers as in the case of [30], in which two different types of continuous differential systems are compared: planar and spatial shaped. In some cases, several differential systems are combined [31].

Contact force recognition and prediction of contact areas are other important aspects to consider when implementing robotic grippers for specific tasks [32,33], in order to prevent breakages, deformations, or release of grasped objects. For this purpose, force sensors can be applied to the robotic fingertip to estimate and regulate the contact force. Among the variety of force sensors, a simple and cost-effective solution to be used for robotic applications is provided by Force Sensing Resistors (FSRs). To obtain correct contact force measurements, these sensors should be characterized, calibrated, and validated on the specific fingertip. Sensor characterization consists of a correlation between the applied force and the output voltage value of the sensor. The purpose of calibration is to obtain an output force value consistent with the applied force when the sensor is implemented on a fingertip. Based on the materials used for the fingertip structure, different calibration procedures must be performed. Thanks to the validation, it is possible to verify the accuracy of the force evaluation. In [34–38], the characterization of FSR was performed but without implementing the sensors in robotic fingers for grasping activities. In [39–41] the FSR was applied to a robotic fingertip to detect the occurrence of contact. In the case of [42,43], the sensor was only directly calibrated on the robotic fingertip.

This paper aims at presenting the study, design, prototyping and testing of a novel underactuated, tendon-driven robotic gripper with four soft monolithic fingers with embedded force sensors. The proposed gripper was named SMART gripper (Soft Manipulator for Adaptive Robotic Tasks). The Force Sensing Resistors (FSRs) integrated into the fingertips enable the contact force recognition and regulation with the aim of avoiding breakages, deformations, and release of the grasped objects. The proposed device introduces novelties regarding design and materials, sensor integration and performance study, as reported and discussed in the next chapters. The new design characteristics that distinguish the SMART gripper from other tendon-driven soft grippers described in the literature are as follows: materials, soft finger structure and properties, differential system. From the sensing point of view, FSRs are integrated in the fingertips using a novel characterization, calibration, and validation method. The paper discusses the key advancements due to the novelties introduced with this project and the impact on industrial applications. In addition, the theoretical and experimental analysis of the finger and fingertip stiffness, trajectories and coefficient of friction during grasping allow to get a huge scenario of the gripper's performance. Finally, the experimental campaigns in real activities give an insight about the use of the developed gripper in industrial applications.This approach represents a comprehensive study of the new proposed gripper.

The paper is structured as follows: Section 2 describes the novelties introduced with the paper regarding gripper design and the materials used; Section 3 describes the force sensor integration, the novelties about characterization, calibration, and validation of the FSRs implemented in the different types of robotic fingers and grasp force control. Section 4 describes all performance studies performed to characterize the gripper: fingertips and fingers stiffness evaluation, experimental trajectories characterization and static friction coefficient experimental evaluation. Section 5 reports all the results and the discussion of the prototyping and characterizations conducted. Section 6 reports on the grasping tests performed. In particular, with regard to the gripper with PETG (Polyethylene Terephthalate Glycol) fingers, drop, deformation, and breakage tests of the gripped objects were carried out, with evaluation of the coefficient of friction of the first contact with two different materials through the use of force sensors. In addition, grip tests were also performed to describe the behavior of the differential system and analyze the distribution of force to the four fingers. Conformal grasping tests were instead carried out on the gripper with TPU (Thermoplastic Polyurethane) fingers. Section 7 reports some industrial applications in which the SMART gripper could be used to exploit its features.

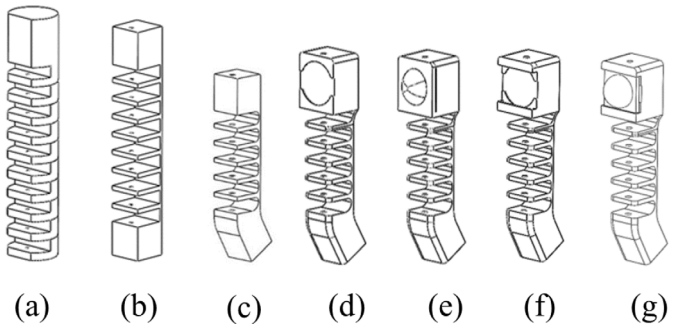


Fig. 1. Finger design evolution.

2. New gripper design and materials

The new robotic fingers were made of flexible plastics, specifically PETG (Polyethylene Terephthalate Glycol) and TPU (Thermoplastic Polyurethane) and the two materials were compared in terms of stiffness, trajectories of the fingertip and contact forces. Few previous works have shown monolithic soft tendon driver fingers made entirely of PETG or TPU. The monolithic structure allows manufacturing both rigid phalanges and flexible joints from one material and this minimizes the complexity of the printing process and the gripper assembly. The choice of these two different materials stems from the desire to study the behavior of the same gripper, composed of fingers with different elastic properties and applicable to different types of gripping. Among the materials commonly used for 3D printing [44], PETG and TPU appear to be the most suitable due to their characteristics, including reduced brittleness, flexibility, impact resistance, and food safety for possible applications in the food and beverage industry. With the aim of performing soft grasps, the finger design evolution included the implementation of a hemispherical silicone fingertip. The initial conceptualization of the new fingers was inspired by the objective of creating an underactuated gripper with soft monolithic fingers and a tendon-driven actuation with the ability to perform a compliant grasping. With the aim to realize this new robotic finger, seven prototypes have been conceived in order to refine the desired final characteristics, and they are represented in Fig. 1.

The key points for optimizing the new soft robotic finger are listed below:

- Monolithic structure
- Easy to integrate with existing palms
- Maximization of finger flexibility
- Optimization of finger dimensions
- Sufficient working volume
- FSR integration on fingertip
- Removable fingertip seat

The characteristics of interest achieved by each prototype are graphically schematized in Fig. 2 by numbering the level of evolution from 0 to 5 in here 5 represents the maximum.

Each version was tested and iteratively modified based on observations during simulation and initial prototyping. Below is a summary of the key changes and motivations driving each transition:

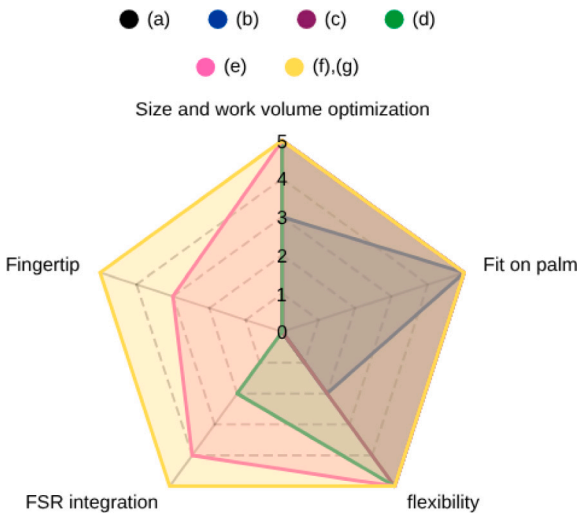


Fig. 2. Graphical representation of the features of interest for each evolutionary stage of the prototype.

- Prototype (a) to (b): Introduced a palm-adapted base and The potentate is thinner to increase flexibility.
- (b) to (c): Optimization of the overall finger size to reduce weight and allow a larger working volume by increasing the distance between the end phalanges.
- (c) to (d): First integration of FSR sensor, requiring mechanical modifications to ensure proper seating and contact surface.
- (d) to (e): Attempt to realize fingertip and finger as a single structure with integration of FSR between the two parts.
- (e) to (f): Introduction of a seat to bind separately made fingertips.
- Prototypes (f) and (g): consist of the same finger structure with and without a silicone fingertip.

The hemispherical fingertip shown in the last design was made of a bi-component soft silicone rubber. A two-part mold was 3D printed to manufacture the silicone fingertips. The liquid silicone mixture was poured into the mold until completely dry.

The models shown in Fig. 1(d) and (g) were used for the experimental calibration and validation of the force sensor and the characterization of contact stiffness for the fingertip. The model shown in Fig. 1(c) was used for the characterization of fingertip stiffness both by experimental and theoretical methods, and for the experimental characterization of the finger trajectories. Only the model in Fig. 1(g) was used for the experimental grasping tests and determination of the coefficient of static friction.

A finger design apparently similar to the proposed one has been used for grippers that exploit pneumatic actuation differently from the tendon-driven one proposed in this paper [18,45]. A robotic finger with a design similar to the one proposed in this paper is reported in [27]. The two differ in the materials used, the fingertip and its sensing. Although the design is similar in that paper the structure is used as a skeleton, surrounded by a silicone cover that forms the contact surface. In this presented work, on the other hand, the only silicone element is in the terminal phalanx, which is the only part of the finger involved in grasping. The target is also different: in [27] a power grasp is mainly performed, while with the SMART gripper a precision grasp is mainly performed.

All the structural elements of the gripper are 3D printed in PETG, but with different filling rates depending on the function to be performed. The schematic of the proposed design of the new robotic gripper is represented in Fig. 3. The box represents the central body of the gripper, to which all the other elements are connected. The top surface of the box acts as the palm of the gripper. In order to have

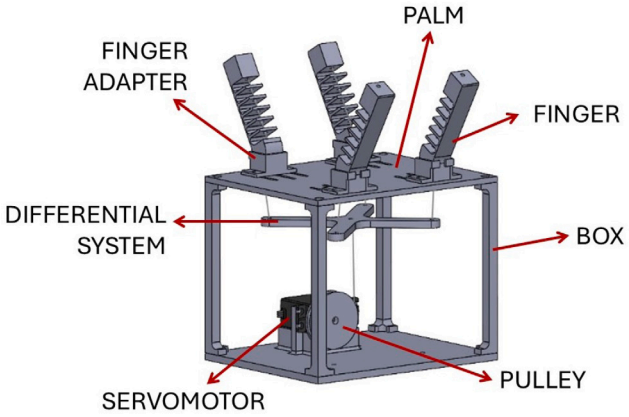


Fig. 3. 3D model of the new gripper with components description.

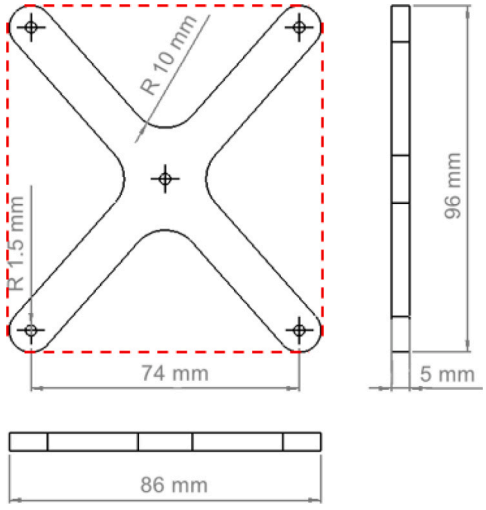


Fig. 4. Differential mechanism dimensions, the red line represents the profile of a rectangular differential used in the experimental phase for comparison.

the maximum adaptability of the gripper to different working volumes, shapes and sizes of the objects, proper slots have been designed on the palm to connect specific adapters that will allow moving and outdistancing the fingers. At the bottom of the box there is a support for the servomotor. A pulley is keyed to the motor shaft to transfer the motion to the fingers through a tendon.

In this work, a new type of seesaw differential system is proposed, extended to the use of four fingers operated by a single actuator. The system is designed as a 3D printed cross with a slot in the center to connect the tendon coming from the servomotor and four slots on the wings to connect the tendons directed to the, as shown in Fig. 4.

The characteristic that distinguishes it from all the cases previously described, for example [30] or even [46] is its structure and flexibility. In fact, the seesaw effect is caused not only by the tilt of the system but also by the bending of the cross wings due to the structure's elasticity. The thicknesses of the system can be properly determined and changed to obtain different and specific transmitted forces, depending on the rigidity. In detail, the model of the differential mechanism looks as in Fig. 5, where one can observe how the force applied to the center is distributed over the 4 wings of the cross: α, β, γ and δ are coefficients representing the distribution of the motor force among the fingers, since in the real case the force will never be uniformly distributed. The sum of these coefficients must equal 1, since all the force delivered by the motor is transmitted to the fingers. The distribution of the motor force among the four wings of the differential system and among

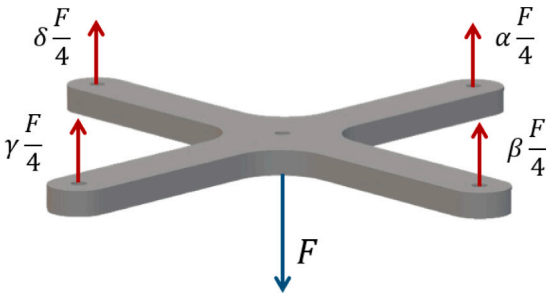


Fig. 5. Force distribution in the differential system.

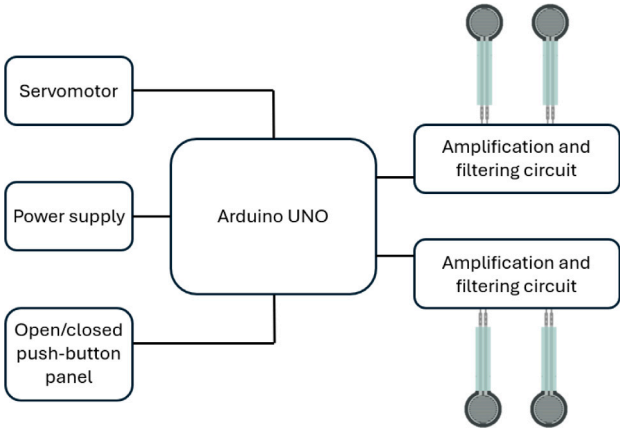


Fig. 6. Power supply and control circuit.

the fingertips was also experimentally determined in different tests to investigate how α, β, γ and δ vary, considering also the influence of the differential's geometric design and material properties, comparing the cross shape differential with a more rigid rectangular one with the same diagonals and thickness (represented by the red dashed line in Fig. 4). Results and discussion of these tests are described in the next paragraphs. Nevertheless, the values that the coefficients α-δ can assume mainly depend on the assembly configuration, the contact points on the objects and the order of contact.

To manage the motor drive, read the contact force through the sensors and control the gripper, an Arduino UNO (Arduino s.r.l., Italy) board [47] was used. Overall, the elements that constitute the power supply and control circuit are: the power supply, the servomotor, a control panel, the Arduino UNO board and sensor signal filters (Fig. 6).

3. Finger sensorization

With regard to the force sensors, they were first characterized and then calibrated by introducing them in four different configurations to enable the real-time measurement and regulation of the contact force exerted by the fingers on the object. In addition, the sensor measurements were also validated. This represents an advancement with respect to other works in literature as previously described. The introduction of the force sensor and its integration into the motor control system via Arduino makes it possible to measure in real time the force exerted by each finger on the object and, at the same time, to set a maximum or minimum threshold to be reached by the fingers during the grasping action, thus enabling grasping without damaging the object.

With the aim of measuring the contact force applied by the fingertip to the grasped object, resistive force sensors were used, and in particular Force-Sensing Resistors (FSRs) (Fig. 7). FSRs are low-cost, thin, and



Fig. 7. FSR.

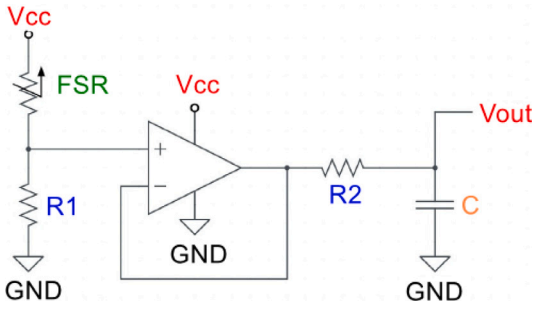


Fig. 8. Sensor connection and conditioning circuit.

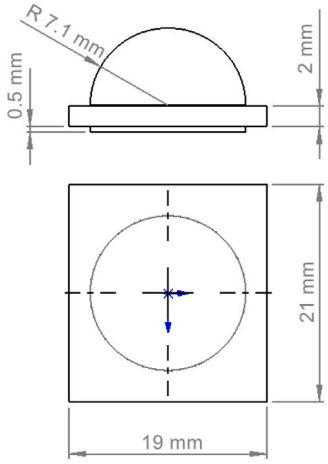


Fig. 9. Dimensioned drawing of the silicone fingertip.

versatile, but suffer from hysteresis, low repeatability, instability, and sensitivity to contact surfaces [34,48].

In order to obtain an accurate output force, the sensor was characterized. After the characterization, the various fingertip types were calibrated considering different shapes and materials. Finally, the validation procedure was performed. The finger designs used during this stage are represented in Fig. 1(d) and (g). The IMADA ZTA-50N digital forced displacement test rig (IMADA instruments, Korea) was used as the reference instrument for force application on the sensor and fingertips. According to the tests, the two main issues that significantly reduce the accuracy of the results are: strong dependence of the output on the compressed surface extent [49], and instability of the output values [50]. To optimize the measurement, the sensor signal was filtered and stabilized, and a proper tip for the digital dynamometer was designed to perfectly fit the sensor shape. The filters are four, one for each force sensor. The signal is first amplified through an operational amplifier and then filtered through an active low-pass filter, as shown in Fig. 8.

In addition, a specific design of the silicone fingertip base was conceived to press the entire FSR's surface during the grasping [32]. Specifically with regard to the construction of the fingertip, a silicone rubber with shore 30 was used, and its dimensions are shown in Fig. 9.

At the bottom there is a thin disk with a diameter equal to the sensing area of the FRS to avoid force dissipation and unwanted resistivity change. The upper part of the fingertip, on the other hand, consists of

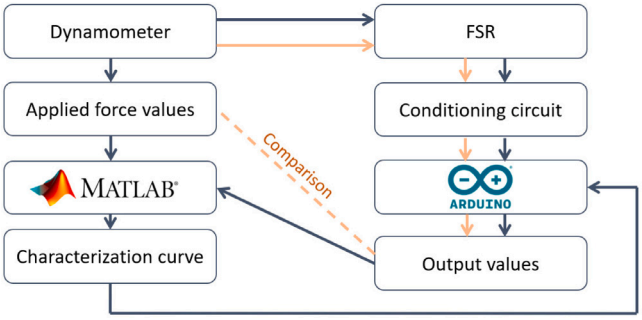


Fig. 10. Characterization, calibration, and validation workflow: The blue line represents the process to derive and implement the sensor characterization and calibration curves while the orange line represents the process followed to perform validation for each step. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

a hemisphere with the same diameter. This shape allows for symmetry in force distribution in case of perfect contact.

For both the sensor characterization and the fingertip calibration and also for the validation, the same method was used (Fig. 10) but using different assemblies of the fingertip. To describe the workflow, the sensor characterization process is taken as an example. A series of known force values was applied to the sensor through the dynamometer. Therefore, two different outputs were obtained: a force–displacement curve as the output from the dynamometer and a samples-bits curve as output from Arduino. As previously mentioned, the Arduino board is connected to the FSR through an amplification and filtering circuit. The acquisition was repeated 10 times to best describe the behavior of the sensor. The obtained outputs were then correlated in MATLAB obtaining a time-independent characterization curve. The equation and coefficients describing the curve were implemented in the acquisition code in Arduino. The characterization process is described by the blue line in Fig. 10. In this way, by repeating the tests with the dynamometer, the force values read by the sensor could be validated (orange line in Fig. 10).

In succession, the procedure workflow was performed by the following steps: characterization of the stand-alone sensor (Fig. 11, Step 1), calibration of the fingertip base (Fig. 11 Step 2) and calibration of the whole finger assembly (Fig. 11, Step 3). This procedure was applied to the fingertips both of PETG and TPU.

It is important to initially characterize the sensor in order to obtain a force value directly at the sensor output. The sensor's bit values are correlated with the corresponding force applied in real time. Having the sensor already characterized means it can be more easily applied to other types of fingers and fingertips.

At the end of each step, the curve obtained was implemented in the Arduino code. Both the characterization and the calibrations were validated by implementing the curves obtained in the various steps and repeating the measurement, verifying that the sensor outputs matched the forces applied by the digital dynamometer and evaluating the error range [32].

The application of this method allowed to obtain a correlation between the force values applied through the dynamometer and the digital output by the sensor that was transformed into a force value. This new step-by-step method for sensor calibration introduces an interesting innovation: it reduces possible errors in force readings due to deformation of the sensor not only as a result of contact with the object, but also as a result of deformation of the elastic materials in contact with the sensor itself. Since it is possible to output the actual value of the force applied from the sensor, it is also possible to control the motor to stop at the desired time using a control system. In the Arduino's motor management code, a control command has been

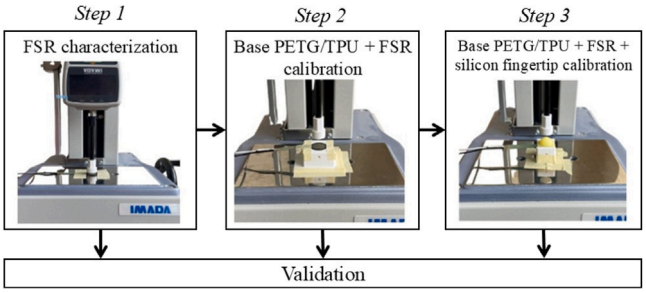


Fig. 11. Step-by-step method and different fingertip assemblies.

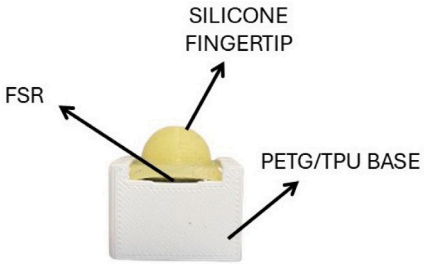


Fig. 12. Fingertip assembly.

implemented which, when a given threshold of contact force is reached for each finger, interrupts the gripper's closing action to ensure a safe grip without damaging the object.

4. Performance study

A further aim of this paper was to study the performance of the new gripper predicting the behavior of the fingers and optimizing the grasping process. For this purpose, an evaluation of the stiffness of the fingertips and fingers was performed. In addition, the trajectories of the fingers were characterized, establishing a correlation between the displacement of the fingertip and the force applied to the tendon. Finally, experimental tests were carried out to validate the effectiveness of the force sensors, ensuring that they are able to prevent objects from falling, breaking, or deforming. Furthermore, the static friction coefficient between the fingertip and the object was evaluated, given the measured contact forces. Grasping Index was also calculated to quantify the grasping performance.

4.1. Fingertips stiffness evaluation

With regard to the characterization of the fingertip contact stiffness, various components and configurations were evaluated: the stand-alone FSR sensor, the PETG and TPU base for the silicone fingertip, the stand-alone silicone fingertip and the two fingertip assemblies in TPU+FSR+silicone and PETG+FSR+silicone (Fig. 12).

The stiffness was characterized using the digital dynamometer to apply specific vertical forces and registering the corresponding displacement. This allowed to determine the force–displacement curves for each considered case. The power law contact model, derived by the Hertz model [51], was considered to evaluate the stiffness:

$$F_n = K\delta^n \tag{1}$$

w here F_n is the normal force, K the stiffness, δ the displacement and n is an exponent determined from the material and the geometrical properties of the local region of the bodies in contact. Knowing the values of force and displacement, it was possible to determine the values of K and n from the fitting power curve of the acquired data [51–53]. The nonlinear least squares approach was used to obtain the various power fitting curves.



Fig. 13. Setup for stiffness experimental characterization of the finger.

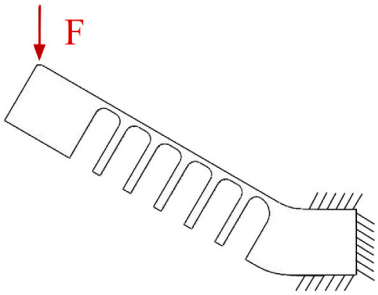


Fig. 14. Scheme with constraints and loads applied for FEM analysis.

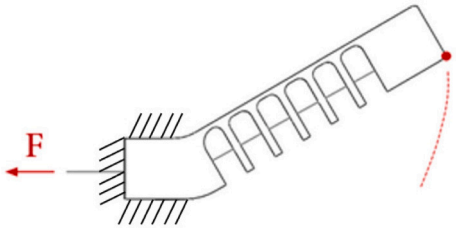


Fig. 15. Scheme of the method applied for experimental trajectory characterization.

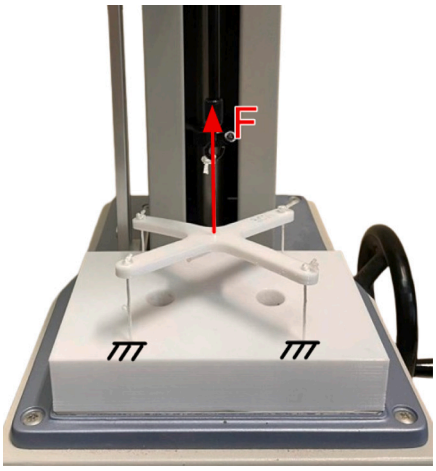


Fig. 16. Setup for determining the force–displacement curve of the differential.

4.2. Finger stiffness characterization

The stiffness of the PETG and TPU fingers was determined using the finger design represented in Fig. 1(c), without integrating the force sensor. The stiffness was evaluated both through an experimental and a theoretical method, and in both cases the finger was schematized as a cantilever beam. A Finite Element Model (FEM) was used for the theoretical study, while the digital dynamometer was used for the experimental tests. During experiments, the finger was constrained to a 3D printed structure and the experimental setup is shown in Fig. 13. The dynamometer applied a series of forces to the fingertip, and these were correlated to the corresponding displacement. This allowed force–displacement curves to be drawn, from which finger stiffness could be derived [54].

The FEM analysis was carried out with the aim of correlating the vertical force applied to the fingertip with the relative displacement in order to derive the stiffness of the finger. Static simulations were performed using a SolidWorks tool, taking care to reproduce loading conditions and constraints comparable to the experimental tests, as shown in Fig. 14. In order to adapt the model to the analyzed fingers, the properties of the materials were obtained from the literature and implemented [55,56].

Based on the experimental results, it was considered appropriate to employ a linear study for PETG and a non-linear one for TPU. In particular, the two-parameter hyperelastic Mooney–Rivlin model was used for TPU [57].

4.3. Experimental trajectories characterization

The trajectories of the fingers were evaluated experimentally in both PETG and TPU, with known values of the force applied to the tendon. For this purpose, the tendon coming out of the base phalanx was connected to the dynamometer's tip. Then, the trajectory of a single marker on the fingertip was traced point by point, recording the value of the applied force for each (Fig. 15). Employing this methodology, the curves of the trajectories were constructed, and the force applied to the tendon at each position was correlated to the vertical displacement.

4.4. Differential structure characterization

In order to characterize the behavior of the cross-shape differential structure, the force–displacement curve was determined applying the dynamometer with a specific setup as reported in Fig. 16. The 4 ends of the cross were constrained while a continuous series of forces was applied to the center of the differential to simulate the gripper actuation.

4.5. Static friction coefficient experimental evaluation

The coefficient of friction can be calculated both theoretically [58] and experimentally using various methods and instruments such as pin on disk machines [59] and inclined planes [60]. A comparison between the theoretical and experimental methods is shown in [61] in the case of rubber. In the case presented in this paper, a simplified experimental method was used, which refers to the inclined plane technique, with the contact surface perpendicular to the supporting plane. Two objects of different materials were gripped and the minimum normal contact force applied by the fingers to prevent the object slipping and release was measured. Since the weight of the object was known, the static coefficient of friction was calculated directly. The measurement was repeated by applying different weights to the object in order to validate the results obtained.

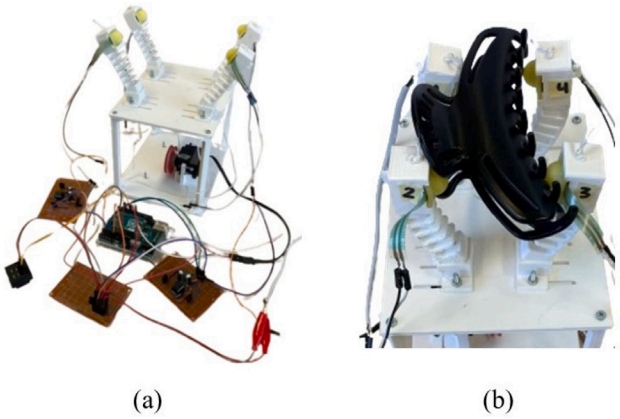


Fig. 17. SMART Gripper: (a) Global gripper representation, (b) Example of gripping a free form object (hair clamp) with PETG fingers.

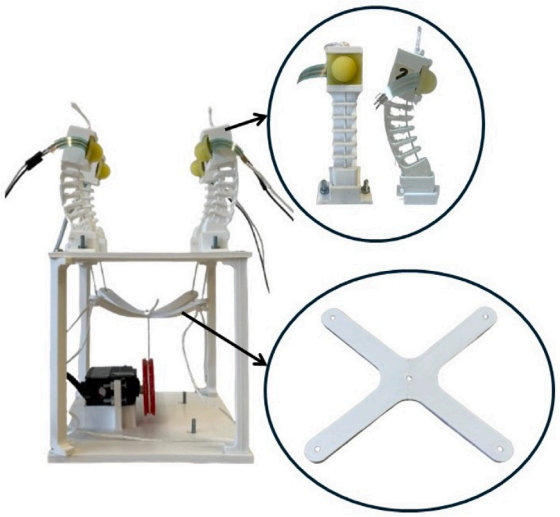


Fig. 18. Gripper prototype detailing fingers and differential system.

5. Results and discussion

5.1. New gripper prototype

After modeling and 3D printing the various components, the gripper was assembled. Fig. 17(a) illustrates the structural configuration of the gripper, comprising the 3D-printed elements, and the circuit component, which incorporates the actuation and sensor management mechanisms described in the preceding section. A compliant grasping is shown in Fig. 17(b).

The presence of the differential allows all fingers to make contact with the freeform object, ensuring a secure grip. Fig. 18 shows a frontal view of the gripper during partial closure of the fingers.

In particular, the two innovative elements that characterize the SMART gripper in its design are highlighted: the monolithic, soft robotics fingers and the deformable cross differential.

5.2. Differential system characterization results

The force–displacement curve of the cross-shaped differential obtained by experimental tests is reported in Fig. 19. This curve allows to correlate the deformation of the differential during test to the force applied by the actuator to get the value of the total force that is distributed among the fingers.

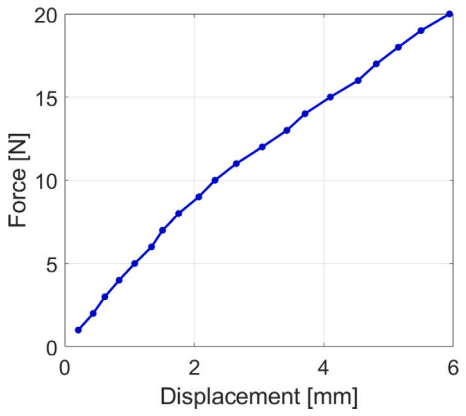


Fig. 19. Force–displacement curve of the differential obtained by experimental test.

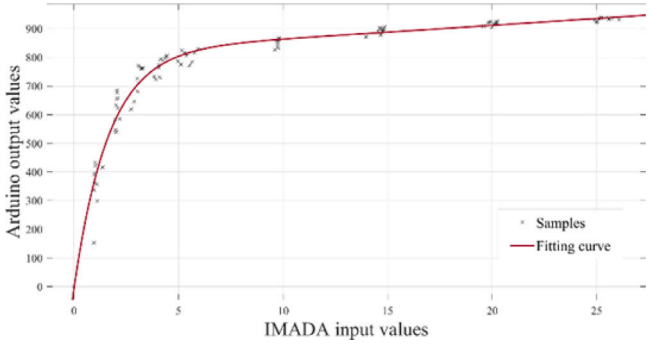


Fig. 20. FSR characterization curve.

Table 1
Parameters describing the goodness of fitting calibration curves.

	R^2	Error range [N]
PETG base + FSR	98%	−0.8 +0.3
TPU base + FSR	99%	−0.3 +0.7
PETG + FSR + sil.	98%	−0.4 +0.9
TPU + FSR + sil.	98%	−0.9 +0.9

5.3. Sensorization results

The characterization curve of the FSR was obtained from the acquisitions made using the setup shown in Fig. 11 (first step). The function that best describes the trend of the samples is a two-term exponential, as shown in Fig. 20. To check the goodness of the fit, the coefficient of determination R^2 was used, which resulted equal to 0.9805. Furthermore, the validation procedure allowed to state that the characterized force value is affected by an average error between −0.8 N and +0.3 N.

The Fig. 21 shows graphically the calibration curves obtained for the different configurations analyzed. For the PETG+FSR configuration of the fingertip, the best-fit curve is represented by a third-degree polynomial function, while in all other cases the curves are fifth-degree polynomial functions.

As with the characterization of the sensor, R^2 and the average error range were determined for the calibration in the different configurations. The obtained values of R^2 and average errors are reported in Table 1 for all the calibrated configurations.

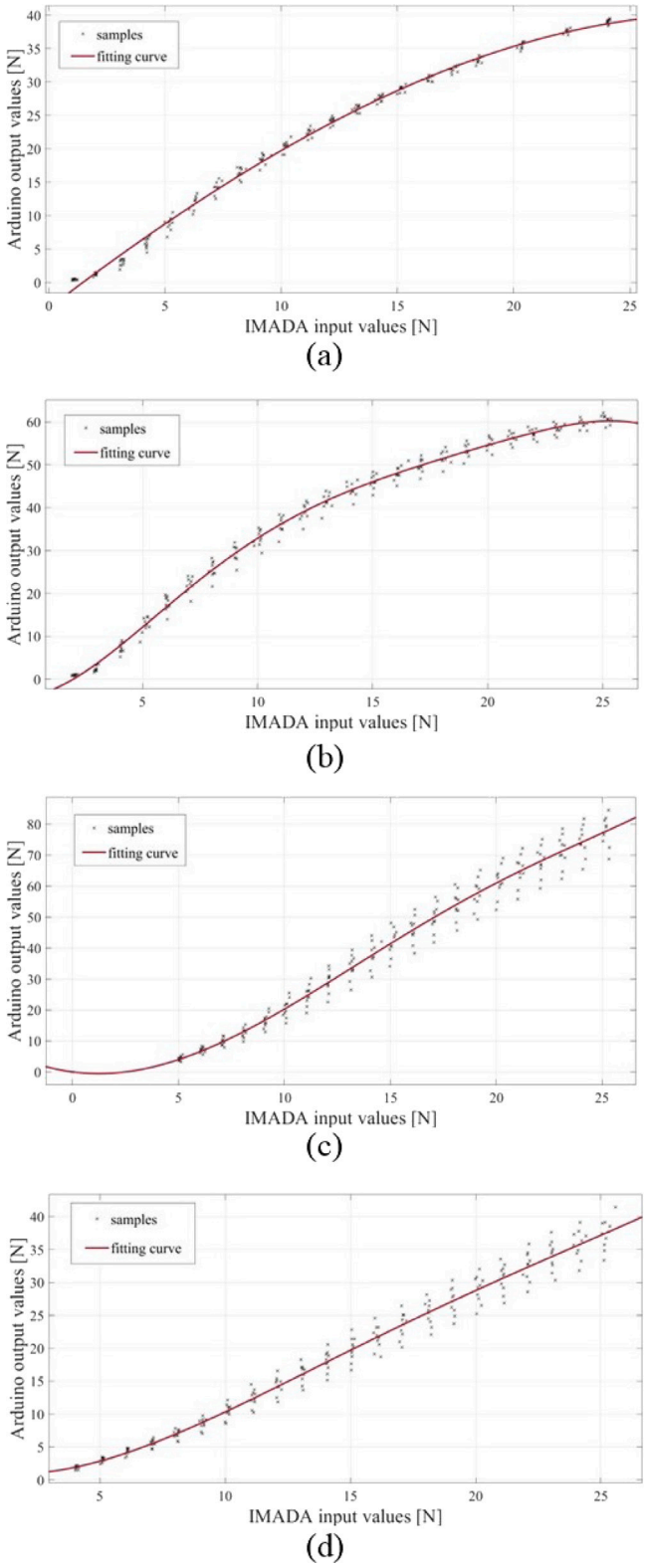


Fig. 21. FSR calibration curves: (a) PETG base + FSR, (b) TPU base + FSR, (c) PETG base + FSR + silicone, (d) TPU base + FSR + silicone.

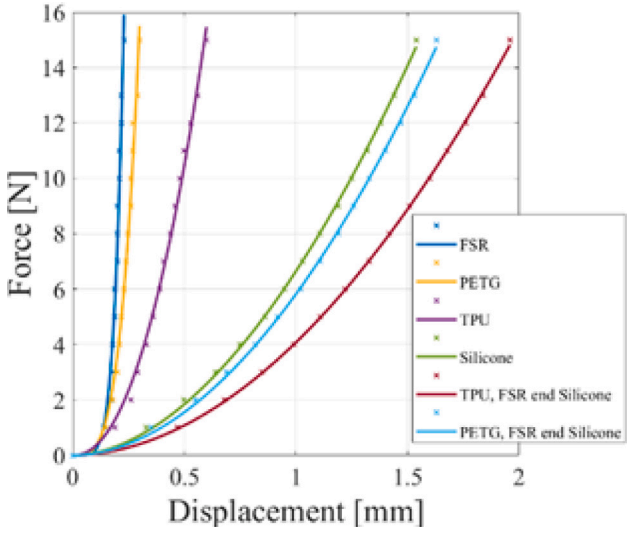


Fig. 22. Fingertip stiffness force–displacement curves.

Table 2
Parameters K and n for each force–displacement curve.

	K [N/mm]	n
FSR	$4.4722 \cdot 10^4$	5.4032
PETG	$1.1780 \cdot 10^3$	3.5967
TPU	46.6579	2.1605
Silicone	96.6386	1.85
FSR + PETG + Silicone	5.8174	1.9034
FSR + TPU + Silicone	4.076	1.9183

5.4. Theoretical and experimental stiffness characterization

5.4.1. Fingertips

By processing the data obtained and applying the power-law contact model, it was possible to determine the force–displacement curves for each type of fingertip studied. Each curve in Fig. 22 is characterized by the parameters K and n which determine its shape. The values are shown in Table 2.

5.4.2. Fingers

The Fig. 23 shows the force–displacement curves obtained experimentally to determine the stiffness of the robotic fingers for both PETG and TPU. Referring to the curve obtained for the PETG finger (Fig. 23(a)), the trend of the samples is described by a fitting curve with a linear trend according to the Hook's law, which is characteristic of elastic materials. From this law, it was possible to determine experimentally the value of the stiffness K of the finger, which was 0.1301 N/mm. A polynomial function was used to describe the step function obtained experimentally with the TPU finger (Fig. 23(b)). The shape of the fitted curve obtained is consistent with the stress–strain curve typical of hyperelastic materials. From the experimental data, the average stiffness K of the TPU finger was derived which was 0.0031 N/mm.

The results of the FEM simulation are represented by a displacement plot in which the color gradient varies according to the extent of the displacement, as shown in Fig. 24. By applying a series of forces, force–displacement curves were again derived for PETG and TPU respectively. By applying a fixed force value as input to the simulation and recording the output value in displacement. The Table 3 compares the force required to apply to the finger in PETG and the finger in TPU to achieve a given displacement.

The theoretical stiffness values obtained from FEM simulations are: 0.1341 N/mm for PETG and 0.0033 N/mm for TPU. The results obtained from both experimental and theoretical characterization were

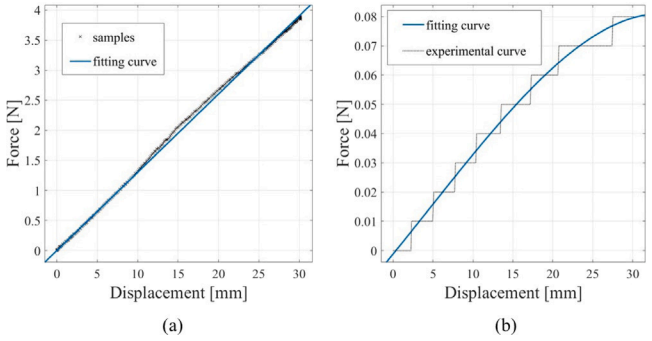


Fig. 23. Force–displacement curve experimentally obtained for: (a) PETG finger, (b) TPU finger.

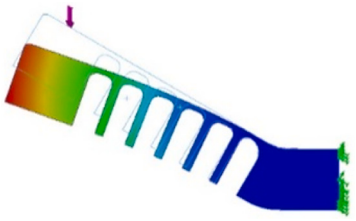


Fig. 24. Fem simulation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Force values required to achieve a certain displacement.

Disp [mm]	Force PETG [N]	Force TPU [N]
3	0.40	0.0106
6	0.80	0.0276
9	1.21	0.0305
12	1.61	0.0401
15	2.01	0.0494
18	2.41	0.0587
21	2.81	0.0681
24	3.22	0.0778

compared for both materials considered. As far as the PETG finger is concerned, it can be seen in Fig. 25(a) that the fit curve obtained experimentally is comparable to that obtained by FEM analysis, and the stiffness K values obtained by the two different methods are also in agreement.

Comparing instead the curves obtained with the TPU fingers (Fig. 25(b)), one can observe that the theoretical curve does not correctly describe the trend of the experimental curve. The curve obtained experimentally using the Mooney–Rivlin two-parameter model is linear due to the low values of the force applied. In fact, when considering a larger range of forces, the curve has a non-linear tendency, typical of hyper-elastic materials. Despite the difference between the two curves, the stiffness values obtained are comparable.

5.5. Experimental trajectory characterization

The outcomes of the finger trajectories are presented in Fig. 26. The Table 4 outlines the forces exerted by the tendon to achieve a specific displacement.

6. Grasping tests and results

One of the objectives of the experimental tests was to assess the effectiveness of various grasping operations. In particular, the ability to perform precision and conformal grasping was investigated for both

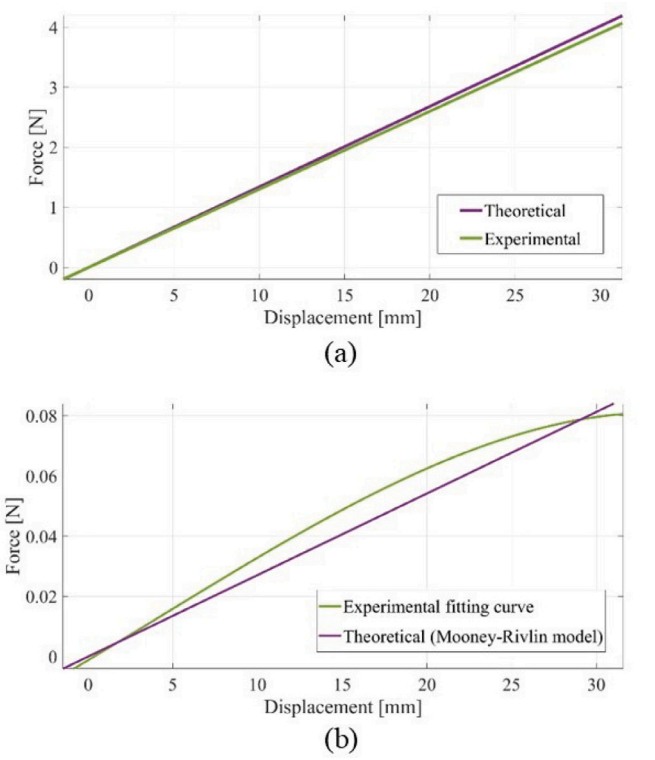


Fig. 25. Comparison of the curves obtained by applying the experimental and theoretical model for: (a) PETG finger, (b) TPU finger.

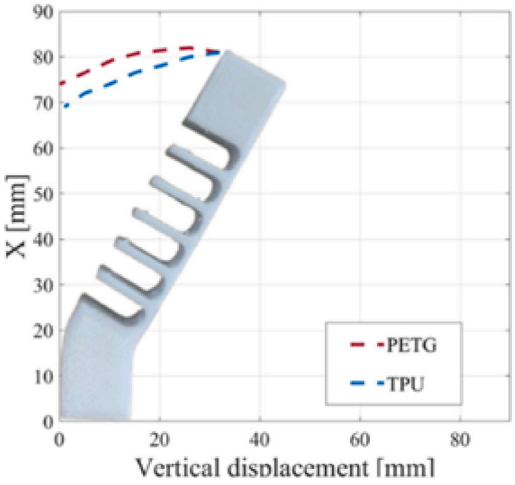


Fig. 26. Trajectory trends.

Table 4
Correlation between the vertical displacement obtained and the force applied to the tendon for each material.

Disp [mm]	5	10	15	20	25
PETG [N]	2.5	5.0	6.0	8.5	10.0
TPU [N]	0.30	0.35	0.40	0.50	0.55

the PETG and TPU fingers. The PETG fingers resulted to be suitable both for precision and conformal grasping, while the TPU fingers, due to their low stiffness, resulted to be ineffective for precision grasping but suitable for conformal grasping, as reported in the next paragraphs. Since most of the tests are aimed at measuring the contact force exerted by the fingertips in precision grasping operations, PETG fingers were

Table 5
Objects properties.

Object	Weight [g]	Material	Fingers
Light bulb	30	Metal, glass	PETG
Bottle 1	367–467	Glass	PETG
Tray	363–463	Plastic	PETG
Can	12	Aluminum	PETG
Empty egg	6	Bio	PETG
Empty cube	73	Metal	TPU
Pencil case	290	Silicone	TPU
Bottle 2	269	Steel	TPU
Stapler	205	Metal, plastic	TPU

used to perform these tests. In addition, both PETG and TPU fingers were used for conformal grasping tests.

Another objective of the tests was to study the contact force during precision grasping of objects. In particular, the overall aim of this test campaign was to regulate the grasping force exerted by the fingers to prevent the object to fall, break and deform. In addition, during the study of the objects' fall, the evaluation of the static friction coefficient between the fingers and various materials was performed. These tests were also useful to analyze how the actuation force is distributed among the four fingers.

In literature, many works present the experimental study of the grasping abilities of the proposed grippers [62–64]. For example, referring to robotic grippers equipped with sensorized fingertips, in [65] the effect of the applied force on a series of objects is evaluated by identifying the minimum force, the maximum force, and the crush force. In [66], hydrogel-based strain and tactile sensors are used to enable force-controlled grasping of fragile or highly deformable objects and to experimentally evaluate both the grasp and the slip of the objects.

During the experimental tests presented in this work, various objects with different weights and characteristics were used, along with robotic fingers made from two different materials. The main characteristics of the experimental setups are shown in Table 5.

6.1. Precision grasping tests

In Fig. 27, some precision grasping tests performed with PETG fingers are reported. Results show how this type of finger is able to grasp and hold different objects just by securing them with the fingertips.

Conversely, Fig. 28, shows some attempts of precision grasping using TPU fingers. One can observe that their low stiffness does not allow to secure the objects just between the fingertips because the entire finger bends during the grasp. Furthermore, contact forces cannot be measured because they are less than the minimum force that the sensor can measure.

For this reason, all the tests focused on precision grasping were performed with the gripper equipped with PETG fingers. In this type of gripping, the only surfaces of the gripper in contact with the grasped object are the fingertips. This allowed to measure the contact force perpendicular to the force sensor on each fingertip.

In all the precision grasping tests, the force applied by the fingers on different objects was between 1.9 and 3.5 N. For lower values, the sensitivity of the FSR given by the calibration curve did not allow the correct measurement of the applied force while force values higher than 3.5 N were not applicable due to the power of the motor and the structure of the differential system. The aim was to determine force levels that could be implemented in the software as a threshold for the force to be applied to prevent the objects being dropped, deformed or broken. The different test setups are summarized in Fig. 27. In addition to these, a test was carried out to investigate the behavior of the differential using a light bulb to assess the following issues: the variation of the parameters α, β, γ and δ during grasping; the relationship between α, β, γ and δ and the input force; the influence of the differential geometric design and material properties (stiffness) on force distribution; how to assess the threshold contact force to apply to preserve the integrity of the grasped objects in all its parts.

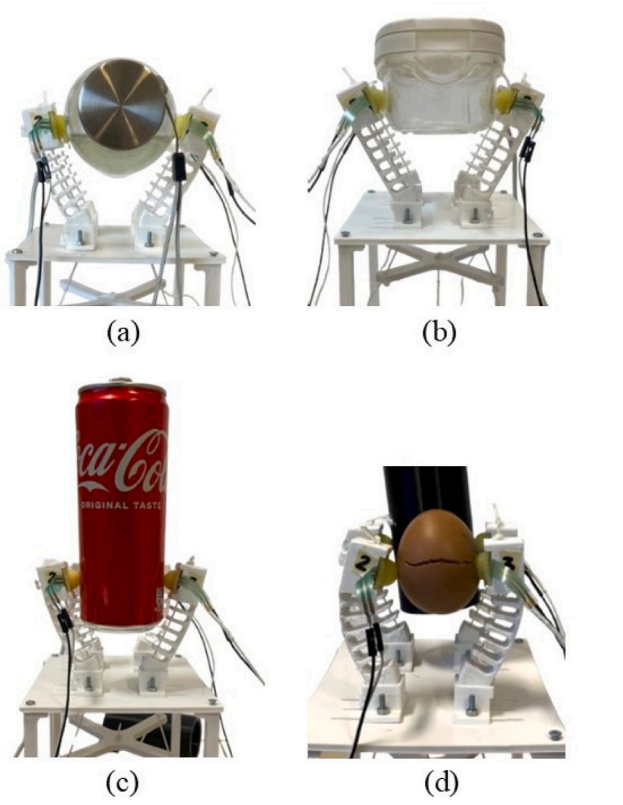


Fig. 27. Experimental tests with PETG fingers: (a), (b) drop (bottle 1 and tray), (c) deformation (can), (d) crack (egg).

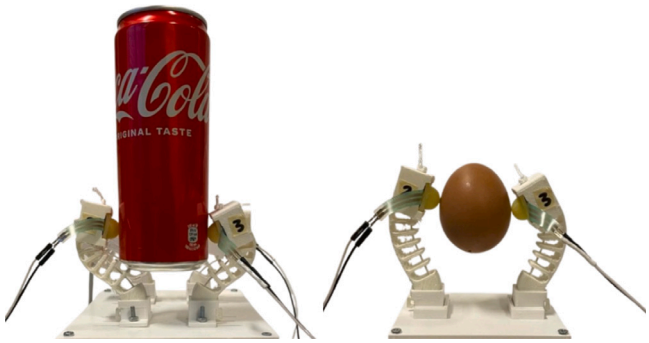


Fig. 28. Precision grasping test with TPU fingers: can and egg.

6.1.1. Ligth bulb test

With the aim of evaluating the behavior of the differential during grasping and the distribution of forces, two grasping tests were carried using a light bulb as target object. Since the light bulb is made of fragile glass on one side and metal on the other, this allowed to understand whether it was possible to obtain a secure grip and determine a threshold value even when grasping heterogeneous objects. In the two experimental tests, two different differential systems were used: the cross-shaped differential and the corresponding rectangular one with the same diagonals and thickness but greater rigidity (Fig. 4). During the tests, the same total contact force exerted by the four fingers on the bulb was applied (as the sum of the 4 forces measured on

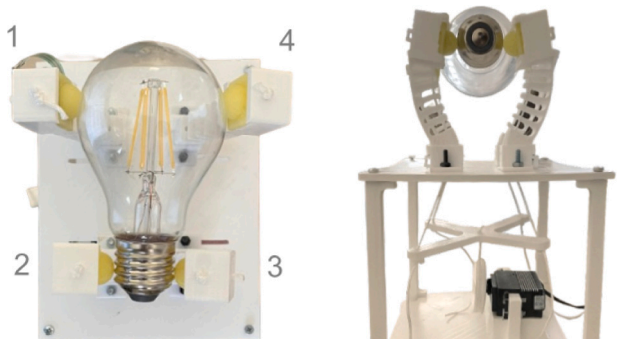


Fig. 29. Light bulb grip test.

the fingertips), considering that the bulb should be held by the four fingertips without damage and drop. The setup of the experiment is represented in Fig. 29, where the number assigned to all the fingers is labeled.

In Fig. 30, the trends and distribution of the contact forces applied by the fingertips on the bulb over time are represented during the grasping with both the differential systems. From these results the following observations can arise. After an initial phase in which the fingers approach the object (phase (I)), the contact phase (II) occurs, which begins with the first fingertip touching the object and ends when all the fingers are in contact with the object (full contact). The order of contact of the fingers can also be highlighted, and one can see that the order of contact can be different test by test. In the contact phase, the cross differential (Fig. 30(a)), takes longer to achieve full contact conditions than the rectangular differential (Fig. 30(b)). This demonstrates the influence of the material properties (rigidity) and of the shape on the grasping action. In fact, the greater deformability of the cross-shape differential implies that the force from the motor is more gradually distributed to the 4 tendons of the fingers. This allows to accomplish delicate grasping of even fragile objects, ensuring that the contact force on the fingertips is applied progressively. In addition, also the cross shape promotes this delicate approach to the object more than a rectangular and more rigid shape/material. In phase (III) (full contact), the force gradually increases up to the limit set for this experiment as the sum of the contact forces, which is 8.7 N.

Another observation that arises is that the motor exerts the same force in both tests. This can be demonstrated considering the total deformation of the two differentials at full grasping and correlating it to the force from the motor that can be determined considering the force–displacement curves (Fig. 19 represents the force–displacement curve for the cross-shaped differential). Fig. 30 also shows that the distribution of force to each fingertip is different using the two differentials. In addition, different tests with the same grasped object and differential would provide different distributions because the contact points on the object can be different for each test and also the contact conditions. This does not represent a problem as the gripper can be programmed to exert a maximum contact force lower than the resistance of the most fragile part of the object but sufficient to hold it preventing drops and damages. In particular, excessive forces on the glass side of the light bulb can be avoided and the grasp can be secure as the force can be set to keep the object stable. In conclusion, the cross-shaped differential system acts as a spring with a specific elastic behavior and stiffness as reported in Fig. 19. This deformability implies that the force from the motor is gradually distributed to the 4 tendons of the fingers up to a specific threshold that is established to avoid drops and damages to the object. This allows to accomplish delicate grasping of even fragile and heterogeneous objects ensuring that the contact force on the fingertips is applied progressively.

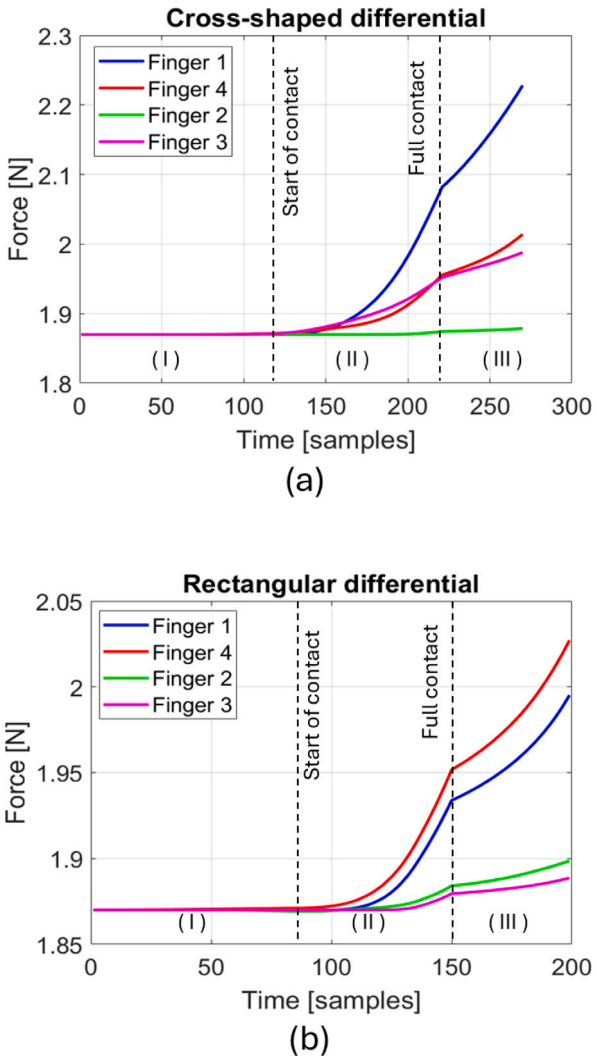


Fig. 30. Force exerted by the four fingertips in time during the bulb grasping: (I) approach, (II) partial contact, (III) full contact; using two different differentials: (a) cross-shaped, (b) rectangular-shaped.

6.1.2. Drop tests

For the drop tests, the minimum force that each finger needed to apply to the object to prevent it from falling was evaluated. The two objects of investigation for the drop tests are a glass bottle and a plastic tray, as in Fig. 27(a) and (b) respectively. This test made it possible to evaluate the coefficients of static friction. To determine the friction coefficients, the contact forces were measured at the moment the objects started slipping between the fingers before falling. In order to validate the friction coefficient, three tests were carried out on the same object, gradually increasing its weight. Table 6 shows the results of the tests on the glass bottle, while Table 7 shows the results of the tests on the plastic tray. The experimentally determined coefficient of static friction is 0.44 for glass and 0.41 for plastic. In fact, these values are consistent with other coefficients attributed to a similar type of couple in contact in the literature [67]. Evaluating the coefficient of friction, even with a limited technique, provides the basis for giving the hand a haptic sense of the friction it is experiencing when grasping.

6.1.3. Deformation tests

For the deformation tests, an object was chosen so that the deformation could be determined both visually and audibly, by detecting the sound the object produces as it deforms. The most suitable object for

Table 6
Glass bottle: contact forces measured at first detachment.

	367 g	417 g	467 g
Finger 1	2,08 N	2,23 N	2,89 N
Finger 2	2,15 N	2,37 N	2,44 N
Finger 3	2,03 N	2,38 N	2,49 N
Finger 4	1,95 N	2,21 N	2,64 N

Table 7
Plastic tray: contact forces measured at first detachment.

	363 g	413 g	463 g
Finger 1	2,29 N	2,54 N	3,10 N
Finger 2	2,27 N	2,46 N	2,52 N
Finger 3	2,20 N	2,46 N	2,62 N
Finger 4	2,19 N	2,32 N	2,74 N

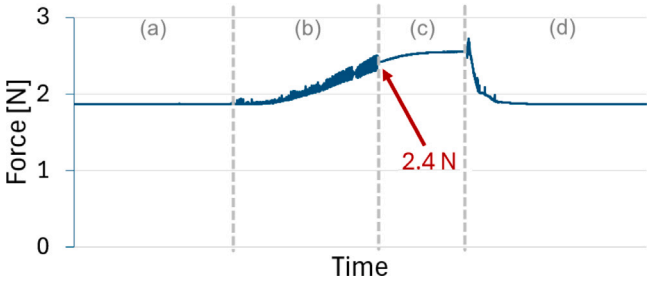


Fig. 31. Force trend over time in the case of deformation testing.

this experiment was a can. Only two fingers were used for grasping the can from as it was placed vertically (Fig. 27(c)). The other two fingers were blocked to allow the differential to work correctly. By determining the force that causes the can to deform, one can establish a threshold of force below which it is possible to perform the gripping action without damaging the object.

The Fig. 31 shows the force trend over time recorded by the sensors in the case of a deformation test. In phase (a), the fingers are activated until contact occurs. In phase (b) of the curve, the gripping action begins with a gradual increase in force and an initial elastic deformation of the can. This part is characterized by an oscillatory force trend due to the resistance that the can itself applies to the sensorized fingertip. When, during the deformation, the can snaps under the force exerted by the two fingertips, the signal stabilizes (phase (c)); one can observe that the corresponding measured force is 2.4 N for each finger. In phase (d) the gripper is released. The peak at the beginning of phase (d) describes the return of the can to its initial undeformed position.

6.1.4. Cracking tests

For the cracking tests, an egg was chosen that had been emptied and cracked beforehand in order to bring the force required to crack it into the range of forces measurable by the sensors. Again, due to the shape of the object, only two fingers were used for gripping, as shown in Fig. 27(d).

Similarly to the previous case mentioned, the threshold for breaking the egg can be determined, which is equal to 2.33 N for each finger. In this case, the force is expected to decrease very rapidly after the break until it reaches zero.

By implementing the values obtained as threshold limits in the motor management, it is possible to perform gripping actions without breaking the object. This is repeatable for any object where the force to be applied to achieve breakage is known.

6.2. Conformal grasping tests

Conformal gripping tests were carried out with the gripper equipped with PETG (Fig. 32) and TPU (Fig. 33) fingers.

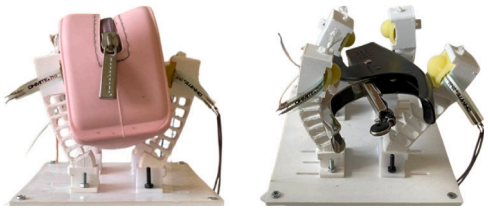


Fig. 32. Conformal grasping test with PETG fingers: pencil bag, stapler.

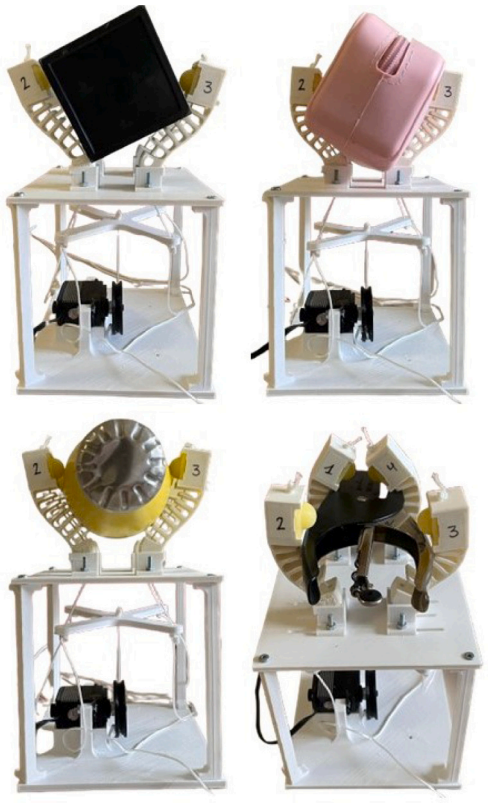


Fig. 33. Conformal grasping tests with TPU fingers: empty cube, pencil bag, bottle 2, stapler.

Both materials are capable of allowing conformal grasping operations. The greater the elasticity of the finger, the greater its ability to adapt to the shape of the object being grasped. For this reason, TPU fingers perform better than PETG fingers in wrapping the different objects even if they can exert lower maximum forces than the PETG finger.

Therefore, for conformal grasping of very light and geometrically complex objects the TPU fingers represent the best solution, while for heavier regular or freeform objects that need to be manipulated by the fingertips or wrapped to perform power grasping, the use of PETG fingers represent the best approach. In conclusion, the PETG fingers can cover also the field of application of the TPU fingers.

7. Industrial applications

In various types of industrial applications, grippers with flexible fingers made of PETG and TPU can offer numerous advantages, particularly in assembly lines for pick-and-place operations, as well as in logistics and adaptive handling systems [68]. The SMART gripper, by combining an underactuated mechanism with force sensors, enables the effective grasping of fragile and deformable objects, ensuring precision

and minimizing the risk of damage, as demonstrated by the experimental tests conducted. Among the various industrial sectors that can benefit from the use of this type of gripper is the food industry. In this field it is important that grippers are able to safely grip food and glass or plastic cases without damaging the products. PETG is an ideal material for food handling as it is food-safe [69].

8. Conclusions

This work presented a comprehensive study of a new under-actuated robotic gripper with soft fingers made of PETG and TPU. In particular, the work began from the analysis of the state of the art and evaluating what was new with the new SMART gripper. The various stages of finger modeling and prototyping were analyzed, as well as the other components of the gripper, all of which were realized using rapid prototyping technologies. With the aim of studying the behavior of the fingers and predicting their function during bending, their stiffness and trajectories were characterized. The fingertips were also modeled in order to evaluate the contact forces transmitted by the fingers to the gripped object, and were found to be ideal for the implementation of FSR. The employment of low-cost force-sensing resistors mounted on robotic fingers has the aim to optimize the grasping of objects for specific tasks. In order to employ them functionally, it is necessary to characterize the force sensors. The experimental activity involved the calibration of the sensor and the definition of a contact model. This procedure will allow the designing of grippers with controlled gripping forces for various industrial applications. The implementation of the gripper was followed by experimental precision grasping tests using PETG fingers and conformal grasping with both PETG and TPU fingers. Using the PETG fingers implemented with the force sensors, it was possible to determine threshold values that could be entered into the motor management code to prevent damages, drop, or cracks to the gripped objects. The static friction coefficients were also evaluated for two different materials (glass and plastic) in contact with the silicone fingertip. These results will also be useful for future directions to integrate the common indices used for grasping evaluation, including the Grasping Index.

CRediT authorship contribution statement

Virginia Burini: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Silvia Logozzo:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Maria Cristina Valigi:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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